

Air Separation Unit

E-Catalogue

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Breathing Life: Your Oxygen Plant Partner

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CRYO CORP O2 LLP

**YOUR GLOBAL
PARTNER IN
INDUSTRIAL GAS
SOLUTIONS**

**Empowering
Industries
Worldwide with
Cutting-Edge Gas
Generation &
Support**

CRYOCORP O2 LLP is a trusted name in the global industrial gas ecosystem, offering a complete range of machinery, spares, accessories, and ancillaries for Medical and Industrial Oxygen, Nitrogen, and Acetylene Plants, along with Bottling Units for Oxygen, Nitrogen, and Argon.

With over 27 years of expertise in the Indian gas industry, the company is helmed by Mr. Ashish Goyal (Managing Director) and backed by a team of seasoned professionals committed to delivering quality, reliability, and long-term performance.

Our solutions are tailored for businesses that value operational efficiency, technical trust, and sustainable growth—earning us recognition as a dependable partner for turnkey gas plant requirements across the globe.

Our clientele spans a dynamic range—from industrial gas plant owners, procurement heads, and plant technicians, to government entities and international export partners with active deployments and AMC/CMC service footprints across Africa, South America, the Middle East, and Southeast Asia.

We specialize in turnkey support for ASU oxygen/nitrogen plants, PSA oxygen & nitrogen generators, acetylene plants, and bottling facilities for gases.

Whether you're building from scratch or upgrading an existing setup, our team ensures uninterrupted performance, high gas purity, and total technical confidence.



We proudly serve clients managing 60–1,000 m³/hr capacities, including notable public sector units like IFFCO and major steel plants. Our expertise also extends to cryogenic liquid storage tanks, manifold lines, CNG hydro-testing machines, and green hydrogen systems—solidifying CRYO CORP's position as a complete infrastructure partner for businesses that demand scale, reliability, and global compliance.

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NEW ASU PLANT

High-Purity Cryogenic Gas Generation Through Air Separation Technology

CRYOCORP's Air Separation Units (ASUs) are fully integrated cryogenic plants that extract high-purity Oxygen and Nitrogen from atmospheric air using ultra-low temperature distillation. Engineered for continuous, high-volume production, they are optimized for long-term, high-volume production sectors like steel, glass, refineries, chemicals, and healthcare. Designed for continuous operation and energy efficiency, these plants offer both liquid and gaseous outputs for diverse applications.

99.99 %

Oxygen purity up to 99.99% for industrial and medical use.

10,000+

Scalable output up to 10,000 Nm³/hr with modular customization.

24/7

Continuous, unmanned operation with automated PLC/SCADA control.

Model	O ₂ Output	N ₂ Output	Purity (O ₂ /N ₂)
ASU-100	100 Nm ³ /hr	200 Nm ³ /hr	99.6% / 99.999%
ASU-500	500 Nm ³ /hr	1000 Nm ³ /hr	99.7% / 99.999%
ASU-1000+	Custom	Custom	Upto 99.7% / 99.999%

Designed for performance, built for trust.

CRYO CORP's ASU plants combine cryogenic precision with intelligent automation, offering seamless dual gas output. The modular design supports scalable capacities, while PLC/SCADA systems ensure real-time monitoring, safety, and control.

Why ASU from CRYOCORP?

- 1. Energy-efficient and continuous operation.
- 2. Flexible output: gas and/or liquid.
- 3. Compliant with ISO, CE, PESO, and GMP standards.
- 4. Designed for harsh Indian industrial conditions.
- 5. Fully automated with remote monitoring capabilities.
- 6. Long service life with CryoCorp installation, commissioning, and AMC support.

Applications

- 1. Steel and Metal Processing Plants
- 2. Medical Oxygen Supply & Bottling Units
- 3. Chemical & Petrochemical Industries
- 4. Glass Manufacturing Units
- 5. Semiconductor and Electronics Plants
- 6. Food Freezing & Inerting Systems



Typical System Components

- 1. Electrical Panel
- 2. Air Compressor
- 3. After Cooler And Cascade Cooler
- 4. Chilling Unit & Chilling Tank
- 5. Dust Filter ,Oil Absorber & Moisture Separator
- 6. Molecular Sieve Battery
- 7. Air Separation Column
- 8. Cryogenic Expansion Engine
- 9. Liquid Oxygen/Nitrogen Pump
- 10. Gas Manifolds & Filling Station

ASU Accessories And Ancillaries

- 1. Industrial Oxygen, Nitrogen and Acetylene Cylinders
- 2. Valve Guard for cylinders
- 3. Cylinder Valves
- 4. On-line analyzers
- 5. Cylinder Hydro Testing Machine
- 6. Porta-Cryo
- 7. Electrical Panel
- 8. Liquid Oxygen and Nitrogen Portable Tanks
- 9. Liquid Oxygen / Nitrogen / Argon Manifold to fill Cylinders

NEW ACETYLENE PLANT

High-Efficiency Calcium Carbide-Based Acetylene Gas Generation System

CRYOCORP O2 LLP offers robust, industrial-grade Acetylene Plants designed for the safe and continuous generation of acetylene gas (C_2H_2) through a controlled chemical reaction between calcium carbide and water. These plants are ideal for use in metal cutting, welding, heat treatment, and laboratory applications, where acetylene's high flame temperature and versatility make it a preferred fuel gas. Built with superior safety, automation, and material quality, our acetylene plants are available in a range of capacities and configurations, conforming to PESO guidelines and global industry standards.

Working Principle:

Acetylene is generated by feeding lump-grade calcium carbide (CaC_2) into a water-filled generator, where a chemical reaction produces acetylene gas and slaked lime ($Ca(OH)_2$). The raw gas is then passed through purifiers, coolers, and driers to remove moisture, impurities, and phosphorus compounds—resulting in pure, safe-to-use acetylene gas for bottling or direct consumption.

Key Features:

- 1. Skid-Mounted Design: Compact layout, pre-piped and pre-wired for easy installation
- 2. Automatic Carbide Feeding System: Ensures consistent generation with minimal operator effort
- 3. Integrated Safety Valves & Flashback Arrestors: Prevents reverse flow and gas hazards
- 4. High-Efficiency Gas Purifiers & Dryers: Ensures acetylene purity up to 98–99%
- 5. Low Power Consumption: Optimized water flow and carbide utilization
- 6. Flameproof Electrical Controls: For safe operation in hazardous areas
- 7. Optional Cylinder Filling Manifold: Available with pressure control and backflow protection.



Applications:

- 1. Metal Fabrication & Cutting Units
- 2. Welding Workshops (Oxy-Acetylene Systems)
- 3. Automotive & Railcar Maintenance Depots
- 4. Foundries & Steel Workshops
- 5. Chemical Synthesis & Research Labs



Technical Specifications:

MODEL	CAPACITY	PURITY	OPERATION PRESSURE	COOLING SYSTEM
ACP-15	15 m³/hr	98% – 99%	Up to 1.5 bar (adjustable)	Water-cooled / Air-cooled
ACP-25	25 m³/hr	98% – 99%	Up to 1.5 bar	Water-cooled
ACP-50+	Custom sizes	98% – 99%	As per client requirement	Custom

Built for performance, backed for life — CRYOCORP's Acetylene Plants offer safe, high-yield operation with minimal fuss and maximum uptime. From *rugged, corrosion-resistant construction to full lifecycle support including installation, training, spares, and AMCs* — we don't just build plants, we build lasting trust across India and beyond.

Second-hand ASU Plants Refurbishment & Relocation

Service Overview



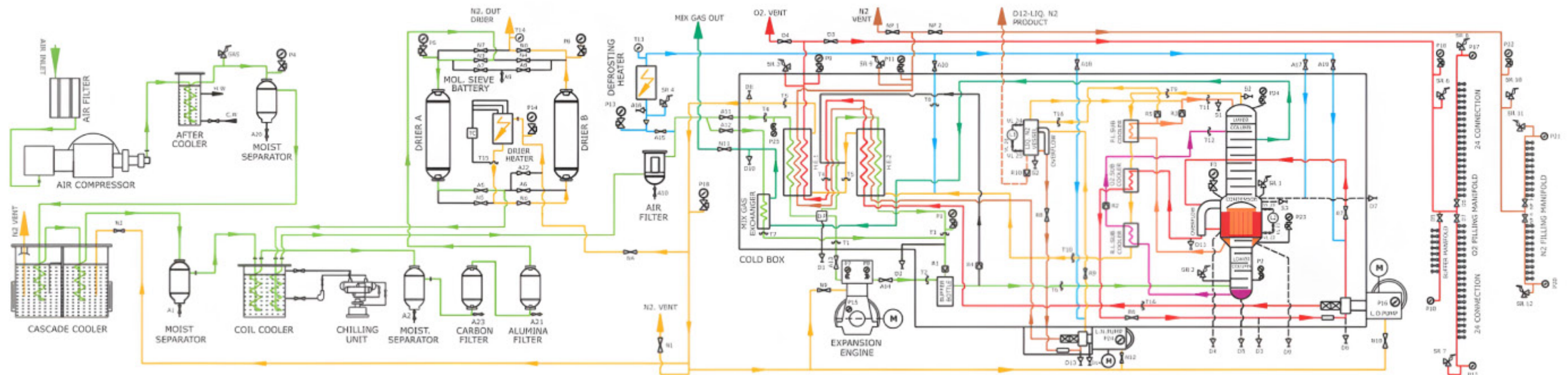
- Applications:
- 1. Steel and Metal Industries
 - 2. Glass and Ceramics Manufacturing
 - 3. Medical Oxygen Supply Units
 - 4. Industrial Gas Refilling Stations
 - 5. Chemical and Petrochemical Plants



Choosing CRYOCORP means partnering with a team that delivers more than machinery — we deliver outcomes. With end-to-end project execution, in-house technical expertise, and proven success across India, Africa, and the Middle East, we offer world-class reliability at a fraction of the cost. Whether it's a new build or a plant relocation, CryoCorp is where precision meets performance.

ASU MACHINERY

Air Separation Units (ASUs) are industrial systems designed to separate atmospheric air into its primary components: oxygen (O_2) & nitrogen (N_2). This is typically achieved through cryogenic distillation, where air is first compressed, purified, and then cooled to cryogenic temperatures to enable liquefaction and separation.



01. Air Compressor & Spares

Precision-Engineered for Continuous Industrial Gas Generation

At CRYOCORP O2 LLP, we understand that a compressor is the powerhouse behind your ASU's performance. Our industrial compressors are purpose-built for 24/7 gas (oxygen, or nitrogen) generation, engineered to deliver uninterrupted reliability, energy efficiency, and clean output. Designed for rugged environments and seamless integration, these systems are tailored to meet the demanding needs of both Indian and international plants..



Technical Specifications (Typical Range):

- 1. Capacity: 100 Nm³/hr to 10,000 Nm³/hr
- 2. Working Pressure: 7 to 35 bar (g)
- 3. Compressor Type: Screw / Reciprocating
- 4. Drive System: Electric Motor / VFD-Driven
- 5. Cooling Type: Air-cooled or Water-cooled
- 6. Noise Level: ≤ 75 dB(A)

Key Features:

- 1. **Oil-Free or Lubricated Flexibility:**
Choose ultra-clean screw compressors or robust rotary/reciprocating systems based on your process needs.
- 2. **High Compression Efficiency:**
Multi-stage systems with intercooling optimize power consumption and thermal stability.
- 3. **Industrial-Grade Durability:**
Vibration-free, corrosion-resistant construction designed for long hours and harsh conditions.
- 4. **Smart Automation:**
PLC-integrated controls and touchscreen HMIs enable real-time monitoring and intelligent load management.
- 5. **Maintenance Made Simple:**
Accessible filters and smart layout reduce downtime and streamline routine servicing.

Applications in ASU's:

- 1. Feeding compressed air into the molecular sieve bed for purification
- 2. Supplying air to the main heat exchanger and expansion turbine for cryogenic separation Essential for PSA/VSA systems
- 3. where stable inlet pressure is critical

Spares for air compressor



- 01. Suction Filter
- 02. Suction Discharge Valve
- 03. Gaskets
- 04. Tube Bundles
- 05. Drain Valve
- 06. Piston Rings
- 07. Gland Packing

Why Choose CRYOCORP?

- 1. Tailored solutions based on plant size and gas purity needs
- 2. Proven performance in oxygen, nitrogen, and argon production
- 3. Backed by nationwide service support and OEM spare availability



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02. After Cooler & Cascade Cooler & Spares

Critical Heat Removal Systems for Enhanced Air Purity & Process Efficiency

In an Air Separation Unit (ASU), the After Cooler and Cascade Cooler play a vital role in conditioning compressed air before it undergoes purification and cryogenic separation. These heat exchangers are designed to efficiently remove residual heat and moisture, thus improving the overall effectiveness, safety, and energy efficiency of the separation process.

1. After Cooler

The Primary Stage Cooling for Compressed Air is installed immediately downstream of the air compressor. Its main function is to reduce the temperature of hot compressed air, condensing out the bulk of moisture and oil vapors that could otherwise harm downstream equipment like molecular sieves or heat exchangers.

Key Features:

- 1. High Thermal Conductivity Tubes: seamless ms pipes or stain less steel
- 2. Air-Cooled or Water-Cooled Options: Configurable based on plant design and ambient conditions.
- 3. Integrated Moisture Separator: Automatically drains condensed water, ensuring dry air supply.
- 4. Robust Construction: Corrosion-resistant materials ensure long service life in industrial settings.



Technical Parameters:

- 1. Inlet Temperature: Up to 120°C
- 2. Outlet Temperature: Typically 35–40°C
- 3. Cooling Medium: Ambient air / Cooling water
- 4. Pressure Rating: Up to 35 bar

2. Cascade Cooler

The Cascade Cooler further reduces the air temperature using a multi-stage cooling system—often using chilled water or refrigerant—preparing the air for cryogenic expansion or adsorption-based purification. This stage is critical for achieving the ultra-low dew point and temperature required in high-purity oxygen/nitrogen generation.

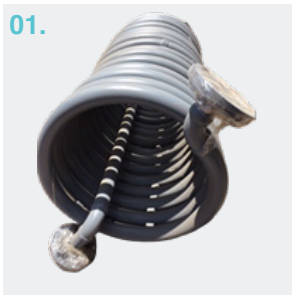


Technical Parameters:

- 1. Cooling Capacity: Custom-designed as per air flow (e.g., 500 Nm³/hr to 10,000 Nm³/hr)
- 2. Temperature Reduction: Down to 5–10°C or lower
- 3. Refrigerants Used: glycol-based chilled water

Key Features:

- 1. Multi-Stage Cooling System: typically double coil system placed side by side in symmetry to enhance cooling effect .
- 2. Designed for Low Dew Points: Helps achieve -60°C to -80°C dew point levels for enhanced air purity.
- 3. Minimal Pressure Drop Design: Engineered for energy efficiency and minimal loss in air pressure.
- 4. Skid-Mounted Options: Compact modular units for easy integration into existing ASUs.



- 01. Vertical Coils
sizes, 1", 1.5", 2" and 2.5" depending on plant capacity
- 02. Side by Side coil for Cascade cooler
size - 1", 1.5", 2" and 2.5" depending on plant capacity
- 03. Complete After Cooler tank with the Coils
- 04. Complete Cascade Cooler tank with s2s coils.



Why Choose CRYOCORP?

- 1. Enhance molecular sieve and cryogenic distillation efficiency
- 2. Improve life span of critical ASU components
- 3. Reduce operating costs by optimizing cooling cycles
- 4. Fully compliant with industrial safety and ISO standards



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03. Chilling Unit & Chilling Tank & Spares

Precision Cooling for Moisture Removal & Gas Purification

In the operation of an Air Separation Unit (ASU), the Chilling Unit and Chilling Tank work together to achieve ultra-low temperatures necessary for removing moisture, hydrocarbons, and other condensable impurities from compressed air. This pre-treatment ensures smooth downstream separation of gases like oxygen and nitrogen, while protecting sensitive components such as molecular sieves and cryogenic columns.

1. Chilling Unit

Refrigeration-Based Cooling System for Air Treatment

The Chilling Unit is a compact refrigeration system designed to lower the temperature of compressed and pre-filtered air to near-zero levels, usually between +2°C and +5°C. By doing so, it condenses residual moisture and hydrocarbons before the air enters the Chilling Tank and, subsequently, the purification section of the ASU.



Technical Specifications:

1. Cooling Capacity: 5 TR to 50 TR (Tons of Refrigeration)
2. Outlet Temperature: 2°C to 5°C
3. Cooling Medium: Refrigerant/Glycol Water
4. Power Supply: 3-phase, 415V, 50Hz

Key Features:

1. Refrigeration Compressor: Scroll or semi-hermetic type with high energy efficiency and stable operation.
2. Plate or Shell-and-Tube Heat Exchanger: Ensures rapid and uniform heat transfer for stable cooling.
3. Digital Control Panel: Precise temperature control with alarms and safety interlocks.
4. Environment-Friendly Refrigerants: Options include R-407C, R-134a, and eco-safe refrigerants.

2. Chilling Tank

Cold Reservoir for Air Residence Time and Moisture Separation

The Chilling Tank complements the Chilling Unit by serving as a reservoir that allows the cooled air to stay in contact long enough for moisture and condensables to settle and separate out. It is typically insulated and designed to maintain the chilled condition of the air, improving the efficiency of dew point reduction before air enters the molecular sieve beds or heat exchangers.



Technical Specifications:

1. Storage Capacity: 300 to 3000 Litres (customized as per ASU capacity)
2. Operating Pressure: 7 to 35 bar
3. Insulation Type: PUF / Glass Wool / Polyurethane Foam
4. Material of Construction: MS / SS304

Key Features:

1. Double-Walled with Insulation: Minimizes thermal losses and maintains low internal temperatures.
2. High-Quality SS Construction: Epoxy quoted schedule 80 pipes to prevent corrosion, suitable for high temperatures
3. Moisture Drain Valves & Sight Glasses: For easy monitoring and periodic drainage of condensate.
4. Skid-Mounted Design: Compact, modular units with easy installation and low maintenance.

Spares for Chilling Unit & Chilling Tank



Why Choose CRYOCORP?

1. Tailored solutions based on plant size and gas purity needs
2. Proven performance in oxygen and nitrogen production
3. Backed by nationwide service support and OEM spare availability

01. Freon Compressor
02. Expansion Valve
03. chilling unit filter
04. Temperature indicator
05. Condenser
06. Outercase
07. Inner coil
08. Outer coil



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04. FRP Cooling Tower & Spares

Reliable Heat Rejection System for Industrial Gas Plants

The FRP (Fiberglass Reinforced Plastic) Cooling Tower is a crucial component in the efficient thermal management of an Air Separation Unit (ASU). It is primarily used to dissipate the heat generated by equipment such as air compressors, chilling units, and intercoolers by cooling the circulating water system. CRYOCORP's high-performance FRP cooling towers are engineered for durability, weather resistance, and efficient heat exchange in demanding industrial environments.

Key Functions:

1. Rejects excess heat from compressed air and chiller systems
2. Ensures consistent operating temperatures for ASU components
3. Improves thermal efficiency and extends equipment lifespan
4. Reduces water and energy consumption with optimized design.



Technical Specifications (Typical Range):

1. Cooling Capacity: 10 TR to 500 TR (Tons of Refrigeration)
2. Flow Rate: 5 m³/hr to 500 m³/hr
3. Material of Construction: FRP body, PVC fills, SS fasteners
4. Temperature Range: Inlet water up to 45°C, outlet down to 30–32°C
5. Fan Type: Axial (direct / belt drive)
6. Structure Type: Round / Square / Cross-flow / Counter-flow
7. Noise Level: ≤ 75 dB(A)

Key Features:

1. *Corrosion-Resistant FRP Casing: Lightweight, UV-resistant, and rust-proof body suitable for all climates.*
2. *Energy-Efficient Fan Motor: Low noise, low power consumption axial flow fan with high air delivery.*
3. *PVC Fills & Splash Bars: High-efficiency fills for enhanced heat transfer and uniform water distribution.*
4. *Compact & Modular Design: Available in square and round configurations for easy installation and scalability.*
5. *Drift Eliminators & Louvers: Minimize water loss and prevent splash-out, improving water retention and air flow.*

Spares for FRP Cooling Tower

01.



02.



03.



04.



Why Choose CRYOCORP?

1. Long service life with minimal maintenance
2. Optimized cooling for high-load ASU systems
3. Resistant to corrosion, scaling, and environmental degradation
4. Quick installation with factory-assembled modules
5. Customizable sizes and designs based on plant layout and load requirements

01. PVC fills
02. Sprinklers for Cooling Tower
03. Cooling Tower fan
04. Cooling tower motor.



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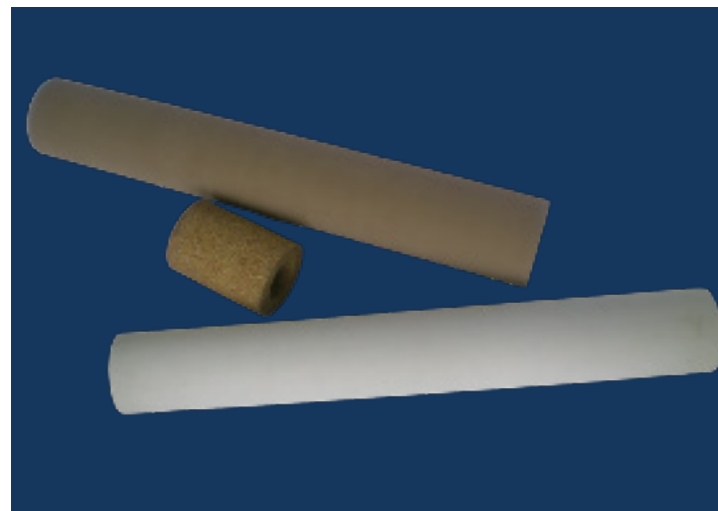
05. Dust Filter & Oil Absorber & Spares

Critical Air Purification Components for Contaminant-Free Gas Separation

In Air Separation Units (ASUs), ensuring the purity of incoming compressed air is essential for achieving high-quality oxygen and nitrogen. The Dust Filter and Oil Absorber are vital elements in the pre-treatment stage, designed to remove solid particles, oil aerosols, and vapors that could damage molecular sieves, clog heat exchangers, or compromise product gas purity. CRYOCORP O2 LLP offers high-efficiency filtration solutions that meet the stringent purity requirements of industrial and medical-grade gas generation.

1. Dust Filter (Pre-Filter / Microfilter) High-Efficiency Particulate Removal

The Dust Filter is engineered to trap and remove fine dust, rust, pollen, carbon particles, and other suspended solids from the compressed air stream. It ensures clean air enters the downstream purification system, thereby improving the performance and longevity of the molecular sieves and cryogenic columns.



Technical Parameters:

1. Operating Pressure: Up to 16 bar
2. Filtration Efficiency: $\geq 99\%$ for particles above 3 microns
3. Flow Capacity: 10 Nm³/hr to 10,000 Nm³/hr

Key Features:

1. Filtration Accuracy: Down to 1–5 microns
2. Low Pressure Drop: Optimized design ensures minimal resistance to airflow
3. Replaceable Filter Elements: Easy maintenance with quick-change cartridge system
4. Robust Housing: Made of aluminum or stainless steel, suitable for high-pressure applications

2. Oil Absorber (Activated Carbon Filter / Coalescing Filter)

Effective Removal of Oil Vapors and Hydrocarbons

The Oil Absorber is a high-performance filtration unit designed to capture oil aerosols, vapors, and trace hydrocarbons that may have bypassed the compressor's oil separator. This stage is essential for ensuring oil-free air reaches the molecular sieve bed and heat exchangers, which are highly sensitive to oil contamination.

Technical Specifications:

1. Oil Vapor Removal: ≤ 0.01 ppm
2. Particle Filtration: Down to 0.01 micron
3. Operating Pressure: Up to 16 bar
4. Flow Rate Range: 20 Nm³/hr to 10,000 Nm³/hr



Key Features:

1. Activated Carbon or Coalescing Media: Ensures deep absorption of oil and VOCs
2. Compact Modular Design: Fits seamlessly into existing ASU air pre-treatment lines
3. High Holding Capacity: Long service intervals with stable filtration performance
4. Corrosion-Resistant Housing: Ensures durability in industrial environments

Spares for Dust Filter & Oil Absorber



01. Activated Carbon Filter
02. Dust Filter
03. Activated Carbon

Why Choose CRYOCORP?

1. Maintains downstream purity and protects critical ASU equipment
2. Increases reliability and efficiency of molecular sieve systems
3. Reduces maintenance costs and unscheduled downtime
4. Custom-engineered solutions based on ASU size and air quality needs



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06. Molecular Sieve Battery & Spares

Advanced Purification System for Ultra-Dry, Hydrocarbon-Free Air

The Molecular Sieve Battery is a core component in the purification section of an Air Separation Unit (ASU), ensuring that the compressed air fed into the cryogenic distillation column is free from moisture, carbon dioxide, and hydrocarbons. These impurities, if not removed, can freeze at cryogenic temperatures and cause blockages or damage within the cold box.

CRYOCORP O2 LLP's molecular sieve systems are engineered for high adsorption efficiency, long operational life, and seamless integration with both cryogenic and non-cryogenic ASUs.



Technical Specifications (Typical Range):

1. Flow Capacity: 200 Nm³/hr to 20,000 Nm³/hr
2. Dew Point After Purification: Up to -80°C
3. Operating Pressure: 6-10 bar
4. Regeneration Gas: Waste Nitrogen or Heated Air
5. Cycle Time: 4-8 hours (customizable)
6. Adsorbent Type: Zeolite 13X / 5A / Activated Alumina

Functionality Overview:

Compressed and pre-cooled air passes through a twin-vessel Molecular Sieve Battery packed with synthetic zeolite adsorbents. One vessel operates in adsorption mode, removing H₂O, CO₂, and trace hydrocarbons, while the other is in regeneration mode, using dry waste nitrogen or heated air to desorb contaminants and restore the media's adsorption capacity. The system operates cyclically to provide continuous purification.

Applications:

1. Pre-treatment before cryogenic air separation
2. Moisture and CO₂ removal to protect heat exchangers and expansion turbines
3. Used in both medical and industrial-grade oxygen/nitrogen generation systems

Key Features:

1. Twin Tower Design: Continuous operation with automatic changeover between adsorption and regeneration.
2. High-Grade Zeolite Media: Specifically selected for deep moisture and CO₂ removal at sub-zero temperatures.
3. Skid-Mounted, Modular Construction: Easy to install and maintain, ideal for compact ASU layouts.
4. PLC/SCADA Controlled System: Fully automated with alarms for pressure, temperature, and valve position.
5. Optimized Thermal Management: Includes heaters and insulation to enhance regeneration efficiency.

Spares for Molecular Sieve Battery



Why Choose CRYOCORP?

1. Prevents freezing of impurities in cryogenic sections
2. Ensures long-term operational reliability of ASU systems
3. Low pressure drop and high adsorption efficiency
4. Reduced downtime through automatic regeneration cycles
5. Available in standard and custom-built configurations

01. Temperature Scanner with RTD Thermocouple
02. Nitrogen Inlet and Outlet Valves
03. Needle Valves for Pressurizing & Depressurizing lines
04. Molecular Sieve & Activated Alumina
05. Electric Heating Element for Battery Heaters.



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07. Air Separation Column & Spares

The Core of Cryogenic Gas Separation – Precision-Engineered for Oxygen, Nitrogen & Argon Recovery

The Air Separation Column is the centerpiece of a cryogenic Air Separation Unit (ASU). This column is where the actual separation of atmospheric air into its primary components — oxygen and nitrogen — takes place using fractional distillation at cryogenic temperatures. Engineered with extreme precision, CRYOCORP O2 LLP's Air Separation Columns are designed to deliver high purity gases, long-term operational reliability, and energy efficiency.

Technical Specifications (Typical Range):

- 1. Production Capacity: 100 Nm³/hr to 10,000 Nm³/hr
- 2. Oxygen Purity: Up to 99.7% (v/v)
- 3. Nitrogen Purity: Up to 99.999% (v/v)
- 4. Operating Pressure:
 - HP Column: ~5–6 bar
 - LP Column: ~1.2–1.5 bar
- 5. Material of Construction: Aluminum alloy (cryogenic grade), SS cladding available

Working Principle:

Once pre-treated and cooled to cryogenic temperatures, the air enters the distillation column system, typically consisting of two vertically stacked columns:
High Pressure (HP) Column:
The incoming liquid-air mixture is partially separated here. Nitrogen-rich vapor rises, and oxygen-rich liquid collects at the bottom.

Low Pressure (LP) Column:

The partially separated gases are further refined under lower pressure to obtain high-purity oxygen at the bottom and high-purity nitrogen at the top.

Depending on design, a third column may be used to extract liquid argon through an intermediate withdrawal point.

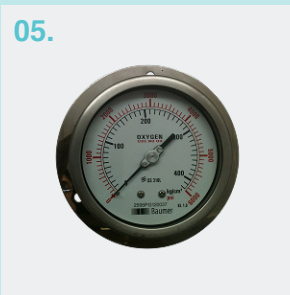
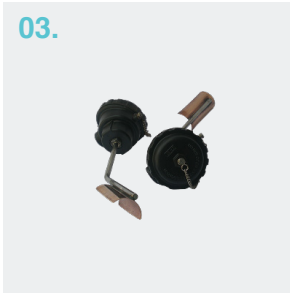
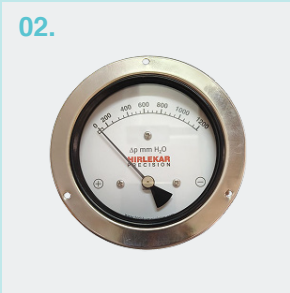
Applications:

- 1. Production of liquid and gaseous oxygen and nitrogen
- 2. Integration with cryogenic storage and cylinder filling systems
- 3. Widely used in steel plants, refineries, healthcare, food processing, and electronics.

Key Features:

- 1. Structured Packing / Sieve Trays: Enhanced mass transfer and energy efficiency.
- 2. Aluminum Construction: Cryo-grade alloy for durability and thermal performance.
- 3. Integrated Reboilers & Condensers: Compact plate-fin design for optimal heat exchange.
- 4. Insulated Cold Box Housing: Preserves cryogenic temps; shields from heat gain.
- 5. Optional Argon Column: Enables high-purity argon recovery for large ASUs.

Spares for Air Separation Column



Why Choose CRYOCORP?

- 1. High recovery rate of all three primary gases
- 2. Superior column internals ensure low energy consumption
- 3. Compact skid-based designs for medium-capacity ASUs
- 4. Designed to international codes (ASME / ISO / CE compliant)
- 5. Supported with complete automation and remote monitoring capability

- 01. Temperature Scanner
- 02. Liquid Level Gauge
- 03. RTD Sensor
- 04. isolation valves
- 05. Pressure Gauges
- 06. Expansion Valves
- 07. Perlite Powder
- 08. Safety Valves for top and Bottom Columns.



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08. Expansion Engine & Spares

Cryogenic Cooling Powerhouse – Enabling Efficient Gas Separation

The Expansion Engine, also known as an Expander, is a critical component in cryogenic Air Separation Units (ASUs). It plays a vital role in achieving the extremely low temperatures required to liquefy air and separate it into its primary components—oxygen and nitrogen. By expanding high-pressure air to low pressure, the expansion engine generates the cryogenic temperatures necessary for the operation of the distillation column inside the cold box. CRYOCORP O2 LLP's offers high-performance turbo-expanders specifically engineered for energy-efficient, continuous operation in industrial gas plants.

Working Principle:

The expander reduces the temperature of compressed and purified air by allowing it to expand rapidly through a turbine, converting pressure energy into mechanical energy. This isentropic expansion results in a significant drop in temperature, assisting in the liquefaction of air and maintaining low temperatures inside the separation column.



Technical Specifications (Typical Range):

1. Inlet Pressure: 4–6 bar
2. Outlet Pressure: ~1 bar
3. Operating Speed: 40,000 to 100,000 RPM
4. Cooling Output: Sufficient to achieve air temperatures below -170°C
5. Energy Recovery: Optional power recovery turbine available
6. Lubrication: Air-cooled or self-lubricating systems (no oil contamination risk)

Key Features:

1. Precision-Balanced Turbine Rotor: For high-speed, vibration-free rotation and minimal maintenance
2. Oil-Free Bearing Design (Air or Magnetic Bearings): Ensures contamination-free, long-life operation
3. High-Speed Shaft with Generator or Brake: Recovers energy or dissipates it safely depending on configuration
4. Cryogenic-Grade Materials: Stainless steel and aluminum alloys for durability and thermal resistance
5. Fully Enclosed in Cold Box or Standalone: Custom mounting based on ASU design

Applications:

1. Used in medium to large cryogenic air separation plants
2. Supports liquefaction process in oxygen, nitrogen, and argon recovery
3. Essential for achieving and maintaining cryogenic conditions inside the ASU cold box

Spares for Expansion Engine



Why Choose CRYOCORP?

1. Increases ASU efficiency by reducing reliance on mechanical refrigeration
2. Enables faster cooldown and higher production stability
3. Low vibration and maintenance design for continuous duty
4. Energy-efficient with optional power recovery capability
5. Designed for extreme temperature and pressure conditions with high reliability.

01. Valve Block Assembly
02. Solenoid Valve & Magneto Coil
03. Push Pump Assembly
04. Puppet Valve Assembly
05. Piston and Piston Rings
06. Cylinder Head Assembly
07. Crankshaft Assembly & Cross Head
08. Cam Assembly



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09. Liquid Oxygen / Nitrogen Pump & Spares

High-Pressure Cryogenic Pumping for Efficient Gas Delivery

The Liquid Oxygen/Nitrogen (LOX/LIN) Pump is a vital post-separation component in Air Separation Units (ASUs), responsible for pressurizing cryogenically stored oxygen or nitrogen for applications such as cylinder filling, pipeline distribution, or conversion to gaseous form. CRYOCORP O2 LLP offers robust and precision-engineered cryogenic pumps designed for continuous-duty, high-pressure, and zero-contamination applications.

Functionality Overview:

After the air is separated and liquefied inside the ASU, liquid oxygen or nitrogen is stored in vacuum-insulated tanks. The LOX/LIN pump draws this liquid at extremely low temperatures (-183°C for LOX and -196°C for LIN) and pressurizes it up to 200–300 bar, enabling vaporization for gaseous use or direct cylinder filling.



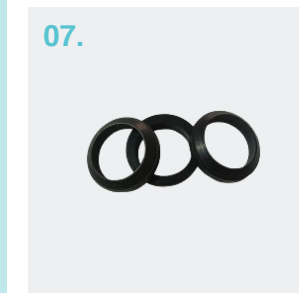
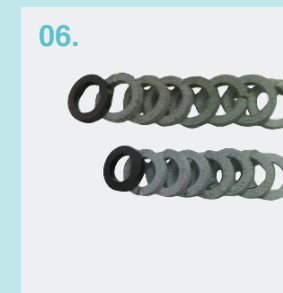
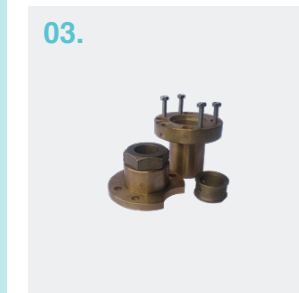
Technical Specifications (Typical Range):

1. Flow Rate: 50 to 1000 liters/hour (customizable)
2. Discharge Pressure: 100 to 300 bar
3. Temperature Range: -196°C to ambient
4. Pump Type: Single-cylinder, double-cylinder or multi-plunger
5. Power Supply: 3-phase, 415V, 50Hz
6. Cooling System: Ambient or optional water-cooled jacket for motor protection

Key Features:

1. High-Pressure Output: Capable of delivering cryogenic liquid at pressures up to 250–300 bar
2. Piston or Reciprocating Design: Ensures reliable and smooth operation under continuous duty
3. Cryogenic-Grade Materials: Stainless steel internals resistant to extreme cold and oxidation
4. Oil-Free Operation: Prevents product contamination—ideal for medical and industrial gases
5. Skid-Mounted & Compact Units: Easily integrable into existing ASU setups
6. Low Vibration and Noise: Designed for quiet, stable performance

Spares for Liquid Oxygen / Nitrogen Pump



01. SS Piston Size 22 mm to 44 mm
02. SS Liner Size 22 mm to 40 mm
03. Safety Valves
04. Piston Ring with Guide
05. Oil Wiper Rings
06. Non Return Valves
07. Gland Packing
08. Gland Box Assembly 28 mm to 34 mm
09. Cross Head
10. Craft Shaft with Connecting Rod
11. Brass Gland Nut & Cylinder Head Nut

Why Choose CRYOCORP?

1. Built for high performance in demanding cryogenic environments
2. Consistent flow with minimal pressure fluctuations
3. Long service life with easy maintenance
4. Low energy consumption due to efficient hydraulic design
5. Ideal for both medical and industrial-grade oxygen/nitrogen distribution



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10. Oxygen / Nitrogen Manifold & Spares

High-Pressure Distribution System for Safe and Efficient Gas Filling

The Oxygen/Nitrogen Manifold is a critical gas handling and distribution system in Air Separation Units (ASUs), especially in setups where oxygen or nitrogen is filled into high-pressure cylinders or supplied to pipelines. CRYOCORP O2 LLP offers precision-engineered manifolds designed for safe, uniform, and high-pressure gas distribution for both medical and industrial applications.



Technical Specifications (Typical Range):

1. Operating Pressure: 150 to 300 bar
2. Outlet Count: 6 to 24 ports (expandable)
3. Material of Construction: SS304, SS316, Brass, or Copper
4. Valve Type: Needle / Ball / Non-return / Check valves
5. End Connection: BSP/NPT threads or as per customer specification
6. Safety Devices: Pressure relief valves, bursting discs (optional)

Functionality Overview:

Once the oxygen or nitrogen is separated and pressurized using cryogenic pumps, the gas is directed into the manifold system. This multi-valve, high-pressure assembly enables simultaneous filling of multiple cylinders or feeding multiple outlets, ensuring balanced gas flow, leak-proof operation, and safe pressure regulation.

Key Features:

1. High-Pressure Rated: Suitable for working pressures up to 200–300 bar
2. Multi-Port Configuration: Available in 6, 10, 12, 24, or custom outlet configurations
3. Non-Return Valves & Isolation Valves: Prevent backflow and allow for flexible operation
4. Durable Construction: SS304/SS316 or copper piping with brass/SS fittings
5. Pressure Gauges & Safety Relief Valves: For monitoring and over-pressure protection
6. Wall-Mounted or Skid-Mounted: Space-efficient design for easy installation

Applications:

1. High-pressure cylinder filling stations
2. Bulk oxygen/nitrogen gas distribution from ASU
3. Medical oxygen bottling and industrial supply
4. Integration with control panels for semi-automatic filling

Spares for Liquid Oxygen / Nitrogen Manifold



01. Cylinder Valves
02. S.S. Spindles
03. Brass Bull Nose & Nuts
04. Spindle Keys
05. Copper Tubes for PigTail Connection
06. Manifold Valves
07. Pressure Gauges
08. Safety valves
09. Angle type Manifold Valve for Outlet of High Purity Oxygen and Nitrogen at 165 kg/cm² pressure.



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Why Choose CRYOCORP?

1. Uniform and simultaneous filling of multiple cylinders
2. Enhanced safety with certified pressure control components
3. Leak-proof joints and long service life
4. Customizable configurations for varied gas plant sizes
5. Easy integration with filling pumps, vaporizers, and storage tanks



ASU Accessories and Ancillaries

Air Separation Units (ASUs) rely on essential accessories and ancillaries to ensure efficient and safe operation. These include air filters, moisture separators, and oil-free booster compressors for clean air and gas pressurization. Control panels manage automation, while cryogenic valves, pipelines, and safety devices handle low-temperature fluids securely. Liquid storage tanks and vaporizers manage gas storage and conversion, and air dryer systems support pneumatic controls. Together, these components enhance reliability and performance of the ASU system

01. Industrial Oxygen, Nitrogen & Acetylene Cylinders

High-Pressure Distribution System for Safe and Efficient Gas Filling

CRYOCORP O2 LLP offers a premium range of Industrial Gas Cylinders designed for the safe storage and transportation of Oxygen, Nitrogen, and Acetylene gases. Manufactured in compliance with international safety and quality standards, these cylinders are widely used across sectors such as metal fabrication, healthcare, chemical processing, and food & beverage industries.



Technical Specifications (Example Range):

Gas	Working Pressure	Water Capacity	Material	Outlet Thread
Oxygen	150–200 bar	10L – 50L	Seamless Steel	Water-cooled / Air-cooled
Nitrogen	150–300 bar	10L – 50L	Seamless Steel	Water-cooled

Product Range:

- 1. Oxygen Cylinders: Used in welding, cutting, combustion, and medical applications
- 2. Nitrogen Cylinders: For inerting, blanketing, purging, and food packaging
- 3. Acetylene Cylinders: Ideal for oxy-acetylene cutting, welding, and heating operations

Key Features:

- 1. High-Pressure Rated: Typically rated for 150 to 300 bar working pressure
- 2. Durable Construction: Seamless steel or alloy cylinders with powder-coated or painted finish
- 3. Standard Valve Configurations: Compatible with BIS / ISO / DOT / TPED valves
- 4. Color-Coded as per Gas Type: For easy identification and compliance with safety codes
- 5. Available in Multiple Sizes: 5L, 10L, 40L, 47L, 50L, and jumbo sizes for bulk usage
- 6. Acetylene Cylinders: Filled with porous mass and acetone for safe acetylene storage

Applications:

- 1. Oxygen: Steel plants, hospitals, welding & cutting, glass manufacturing
- 2. Nitrogen: Electronics, chemical blanketing, food preservation, pharmaceuticals
- 3. Acetylene: Heavy metal cutting, brazing, and repair work in construction and workshops

Why Choose CRYOCORP?

- 1. Certified to IS, ISO, and PESO standards
- 2. Pressure-tested for safety and durability
- 3. Optional accessories: cylinder trolleys, manifold connectors, valve guards
- 4. Available with gas filling, hydro testing, and documentation support
- 5. Fast delivery and after-sales service for industrial and medical sectors

02. Valve Guard for Cylinders

Reliable Protection for High-Pressure Gas Cylinder Valves

The Valve Guard is a critical safety accessory designed to protect the valve assembly of gas cylinders from mechanical damage during handling, transportation, and storage. CRYOCORP O2 LLP offers heavy-duty valve guards engineered to meet industrial safety standards for cylinders containing Oxygen, Nitrogen, Acetylene, Argon, and other industrial gases.



CRYOCORP O2 LLP supplies valve guards individually or bundled with new cylinders, manifolds, and trolleys. Custom branding and laser marking are also available for fleet identification.

Purpose & Functionality:

Cylinder valves are among the most vulnerable components and a common source of gas leaks or accidents if damaged. The valve guard creates a physical barrier around the valve to absorb impacts and prevent direct contact with external surfaces or equipment, ensuring operator safety and regulatory compliance.

Material	Mild Steel / Aluminum Alloy
Finish	Powder Coated / Hot Dip Galvanized
Mounting Type	Threaded / Bolt-On
Applicable Cylinder Size	10L to 50L (custom options available)
Color Options	Black, Blue, Red, Green (or as per gas type)

Applications:

1. Industrial gas cylinder fleets (oxygen, nitrogen, CO₂, argon, acetylene)
2. Medical oxygen cylinders for hospitals and ambulances
3. Gas refilling plants, welding stations, and distribution depots
4. Transportation of high-pressure cylinders in vehicles or trolleys

Key Features:

1. Heavy-Duty Construction: Manufactured from high-strength mild steel or aluminum alloy for durability and impact resistance
2. Corrosion-Resistant Coating: Powder-coated or galvanized finish for enhanced outdoor performance
3. Universal Fit: Compatible with standard 47L, 50L, and jumbo gas cylinders
4. Secure Mounting Design: Bolted or threaded options for tight, tamper-proof fit
5. Open & Closed Type Guards: Available in open-top (ventilated) or closed-top (full dome) styles depending on application
6. Color Customization: Available in gas-specific safety color codes as per ISO / IS standards

Why Choose CRYOCORP?

1. Prevents valve breakage, leakage, and hazardous accidents
2. Enhances safety during cylinder transport and storage
3. Compliant with BIS, PESO, and ISO cylinder safety norms
4. Long service life with minimal maintenance
5. Available in bulk quantities with fast delivery

03. Cylinders Valves

Precision-Engineered Valves for Safe and Controlled Gas Flow

The Cylinder Valve is a critical control component mounted on high-pressure gas cylinders to regulate the release, storage, and refilling of compressed gases such as Oxygen, Nitrogen, Acetylene, Argon, Carbon Dioxide, and more. CRYOCORP O2 LLP offers a wide range of ISI-marked, PESO-approved, and ISO-compliant cylinder valves for both industrial and medical applications.



CRYOCORP O2 LLP ensures that every valve meets rigorous safety and performance benchmarks, offering reliable gas control with long service life. Bulk quantities are available with full technical documentation and compliance support.

Technical Specifications (Example Range):

PARAMETER	SPECIFICATION
Working Pressure	150 to 300 bar
Material	Forged Brass / SS304 / SS316
Connection threads	BSP / NPT / IS:3224 / DIN
Valve Type	Handwheel / Key-operated / Toggle / Pin-type
Certification	ISI Marked, PESO Approved, ISO Compliant

Available Types:

1. Oxygen Cylinder Valves (O₂) – For industrial and medical-grade cylinders
2. Nitrogen Cylinder Valves (N₂) – For inert gas usage in industrial processes
3. Acetylene Cylinder Valves (C₂H₂) – Flame-proof valves with porous filler compatibility
4. CO₂, Argon, Helium, and Mixed Gas Valves – Suitable for food, welding, and lab use

Purpose & Functionality:

Cylinder valves act as the primary control point for filling and withdrawing gas from pressurized cylinders. They are designed to withstand high internal pressure, ensure leak-tight performance, and allow smooth, precise opening and closing under varied operating conditions.

Key Features:

1. High-Pressure Rated: Suitable for up to 300 bar working pressure
2. Material of Construction: Made from forged brass or stainless steel, ensuring strength and corrosion resistance
3. Thread Compatibility: BSP / NPT / ISO / DIN standards based on regional and gas-specific needs
4. Soft Seat or Metal Seat Design: For reliable sealing and long service life
5. Integrated Safety Devices: Options for burst discs, pressure relief valves, or non-return valves
6. Tamper-Proof Design: Security caps, lockable handles, and anti-removal features available

Applications:

1. Gas filling plants and ASUs
2. Welding and cutting operations
3. Medical oxygen supply systems
4. Laboratory and specialty gas handling
5. Industrial cylinder fleets and rental operations

Why Choose CRYOCORP?

1. Manufactured to stringent quality control standards
2. Leak-tested and pressure-tested before dispatch
3. Compatible with international cylinder specifications
4. Available with accessories: valve guards, outlet connectors, and caps
5. Custom engraving or barcoding available for tracking and identification

04. On-Line Gas Analyzers

Real-Time Gas Purity Monitoring for Process Control & Compliance

On-Line Gas Analyzers are essential instruments in Air Separation Units (ASUs) and industrial gas systems, used for continuous monitoring of gas purity and composition. These analyzers ensure that oxygen, nitrogen, argon, or mixed gases meet required specifications before they are stored, distributed, or filled into cylinders. CRYOCORP O2 LLP offers high-accuracy on-line analyzers designed for rugged industrial use, with real-time data output, alarm functions, and seamless integration with automated plant controls.



CRYOCORP O2 LLP provides analyzer systems tailored to your plant configuration, with full commissioning support, calibration services, and AMC options. Available in single or multi-gas models, with remote monitoring capabilities for modern gas plants.

Purpose & Functionality:

Installed directly in the gas output line, an on-line analyzer measures gas purity levels, residual contaminants (like moisture, CO₂, hydrocarbons), and trace gases. These analyzers are critical for maintaining product quality, process safety, and compliance with industrial, medical, and food-grade gas standards.

Applications:

- 1. Continuous purity monitoring in cryogenic ASUs
- 2. Cylinder filling stations (medical & industrial oxygen/nitrogen)
- 3. Process control in food packaging, pharma, steel, and electronics industries
- 4. Safety interlocks for inerting and gas blanketing systems

Why Choose CRYOCORP?

- 1. Improves product quality assurance and regulatory compliance
- 2. Prevents off-spec gas delivery with early alerts
- 3. Reduces manual sampling and downtime
- 4. Long sensor life with minimal recalibration needs
- 5. Compatible with SCADA, DCS, and PLC systems for automation

Technical Specifications (Example Range):

Material	Mild Steel / Aluminum Alloy
Finish	Powder Coated / Hot Dip Galvanized
Mounting Type	Threaded / Bolt-On
Applicable Cylinder Sizes	10L to 50L (custom options available)
Color Options	Black, Blue, Red, Green (or as per gas type)

Key Features:

- 1. Real-Time Measurement: Continuous and precise gas analysis without manual sampling.
- 2. High Accuracy Sensors: Electrochemical, paramagnetic, thermal conductivity, or zirconia-based sensors depending on gas type.
- 3. User-Friendly Interface: Touchscreen display with data logging, USB export, and digital readouts.
- 4. Multi-Gas Configuration: Available for O₂, N₂, Ar, CO, CO₂, CH₂, H₂O, and other trace gases.
- 5. Alarm & Control Integration: High/low setpoint alarms and relay outputs for plant safety interlocks.
- 6. Compact, Wall or Panel Mounted Design: Ideal for control rooms or inline process panels.

05. Cylinder Hydro Testing Machine

Automated Pressure Testing System for Gas Cylinder Safety Assurance

The Cylinder Hydro Testing Machine is a crucial safety and compliance tool used to test the structural integrity of gas cylinders under high pressure. This machine ensures that cylinders storing oxygen, nitrogen, acetylene, and other industrial or medical gases are fit for continued use, in accordance with PESO, ISO, and BIS standards. CRYOCORP O2 LLP offers robust, semi-automatic and fully automatic hydro testing systems designed for precise pressure monitoring, safe operation, and high throughput in gas refilling and requalification centers.



CRYOCORP O2 LLP offers end-to-end solutions including machine supply, installation, operator training, and certification support. Available in standard and custom capacities for high-volume gas cylinder requalification centers.

Technical Specifications (Example Range):

Parameter	Specification
Test Pressure	Up to 350 bar (adjustable)
Cylinder Type	Oxygen, Nitrogen, CO ₂ , Argon, Acetylene
Cylinder Capacity	5L to 80L
No. of Cylinders per Cycle	2–10 (customizable)
Pump Type	Electric or Hydraulic High-Pressure Pump
Control	Manual / Semi-Automatic / PLC Controlled
Safety Feature	Auto-shutoff, Over-pressure alarm, Enclosure lockout

Purpose & Functionality:

During hydrostatic testing, cylinders are filled with water and pressurized to levels significantly higher than their normal working pressure. This checks for leakage, deformation, and metal fatigue. Cylinders that pass the test are certified for further use, while defective ones are safely rejected.

Key Features:

- 1. Multi-Cylinder Testing Bench: Simultaneous testing of 2 to 10 cylinders (expandable)
- 2. High-Pressure Water Pump Unit: Capable of delivering pressure up to 350 bar
- 3. Digital Pressure Gauges: With real-time monitoring and automatic pressure cut-off
- 4. Burst Protection Chambers: Rugged steel enclosure to contain accidental ruptures
- 5. Leak Detection & Deformation Analysis: Auto drain and measurement system available
- 6. Control Panel: PLC or semi-automatic options with pressure cycle programming
- 7. Robust Frame Construction: Powder-coated steel with rust protection and industrial-grade durability

Applications:

- 1. Periodic testing of industrial and medical gas cylinders
- 2. Gas bottling plants, ASU facilities, and authorized requalification centers
- 3. Compliance with PESO, ISO 9809, and other international testing standards
- 4. Ensures safety before refilling and dispatch of gas cylinders

Why Choose CRYOCORP?

- 1. Accurate testing for pressure integrity and wall deformation
- 2. Reduces testing time with batch processing
- 3. Easy maintenance and long service life
- 4. Helps maintain regulatory compliance with minimal manual effort
- 5. Available with full test documentation and certification printing module (optional)

06. Cooling Towers

Efficient Heat Rejection Systems for Industrial Cooling Applications

Cooling Towers are essential components in industrial gas plants, including Air Separation Units (ASUs), where they help dissipate excess heat from equipment like air compressors, chilling systems, and water-cooled machinery. CRYOCORP O2 LLP offers high-performance FRP Cooling Towers designed for maximum thermal efficiency, corrosion resistance, and long operational life.



CRYOCORP O2 LLP provides end-to-end cooling solutions including tower sizing, pump selection, piping layout, and commissioning support. All towers are built to meet IS, ISO, and industry-standard thermal performance norms.

Technical Specifications (Example Range):

Parameter	Specification
Cooling Capacity	10 TR to 500 TR
Flow Rate	5 m³/hr to 500 m³/hr
Inlet Water Temp	Up to 45°C
Outlet Water Temp	30°C – 32°C (typical)
Fan Type	Axial flow (direct or belt drive)
Structure Type	Round or Square, Counter or Cross-flow
Material	FRP Casing, SS Fasteners, PVC Fills

Purpose & Functionality:

Cooling towers work on the principle of evaporative cooling**—hot water from the plant is sprayed inside the tower and cooled by air flow before being recirculated back to the system. This closed-loop process maintains optimal operating temperatures, ensures energy efficiency, and protects equipment from overheating or thermal wear.

Key Features:

- 1. FRP (Fiberglass Reinforced Plastic) Construction: Lightweight, weatherproof, and corrosion-resistant
- 2. Energy-Efficient Axial Fans: Low-noise operation with high air delivery
- 3. PVC Honeycomb Fills: Enhances surface area for better water-air contact and efficient cooling
- 4. Drift Eliminators: Minimizes water loss through air discharge
- 5. Compact & Modular Design: Easy installation, transport, and scalability
- 6. Maintenance-Friendly Access: Simple servicing of nozzles, motors, and fills

Applications:

- 1. Periodic testing of industrial and medical gas cylinders
- 2. Gas bottling plants, ASU facilities, and authorized requalification centers
- 3. Compliance with PESO, ISO 9809, and other international testing standards
- 4. Ensures safety before refilling and dispatch of gas cylinders

Why Choose CRYOCORP?

- 1. Improved equipment performance and energy efficiency
- 2. Long service life in corrosive and humid environments
- 3. Low water and power consumption
- 4. Ready-to-install units with minimal site work
- 5. Customizable for specific flow and cooling requirements

07. Electrical Control Panels

Centralized Power & Automation for Industrial Gas Plant Operations

The Electrical Panel is the command and control hub of any industrial plant, enabling safe, efficient, and automated operation of machinery like compressors, pumps, chillers, vaporizers, and filling systems. CRYOCORP O2 LLP offers custom-engineered electrical panels designed to handle the electrical and automation demands of Air Separation Units (ASUs), acetylene plants, and gas cylinder filling stations. Built with premium-grade switchgear, wiring, and enclosures, our panels ensure reliable operation, simplified maintenance, and long service life in demanding industrial environments.



CryoCorp O2 LLP provides complete design, fabrication, wiring, testing, and commissioning of electrical panels, along with detailed electrical drawings, operating manuals, and support for integration with your plant's automation systems.

Technical Specifications (Example Range):

Parameter	Specification
Voltage Rating	415V AC, 3 Phase, 50Hz (custom on request)
Control Voltage	24V DC / 110V / 230V AC
Current Rating	50A to 1000A (based on plant load)
Panel Types	PCC, MCC, Starter Panels, VFD Panels
Material	CRCA steel, powder-coated or SS (optional)
Cooling	Natural or forced air ventilation
Mounting Type	Floor-standing / Wall-mounted

Purpose & Functionality:

Electrical panels consolidate the control and distribution of power across all plant equipment. They are responsible for starting, stopping, protecting, monitoring, and automating various units in a coordinated manner. Panels can be integrated with PLC/SCADA systems for advanced automation, data logging, and remote monitoring.

Key Features:

1. Modular Design: Panel types include Power Control Center (PCC), Motor Control Center (MCC), and Automation Panels
2. Branded Components: Uses Siemens / Schneider / ABB / L&T switchgear and relays
3. Advanced Protection Systems: MCBs, MCCBs, ELCBs, overload relays, surge arresters
4. Automation Ready: Options for PLC, VFD, HMI, and SCADA integration
5. Weatherproof Enclosures: IP55/IP65 rated panels for indoor or outdoor use
6. Neat Internal Wiring: Color-coded, tagged, and compliant with IS/IEC standards

Applications:

1. Air Separation Units (Cryogenic & PSA Plants)
2. Oxygen/Nitrogen Filling Stations
3. Acetylene Gas Plants
4. Water-cooling, chilling, and vacuum systems
5. Cylinder hydro testing and automation setups

Why Choose CRYOCORP?

1. Centralized and fail-safe operation of plant equipment
2. Fully customizable layouts and wiring schematics
3. Safety interlocks, alarms, and emergency stop provisions
4. Ready for expansion and future automation upgrades
5. Designed, assembled, and tested in compliance with IS/IEC norms

08. Porta Cryo – Portable Cryogenic Liquid Cylinder

Mobile Storage & Transport Solution for Liquid Oxygen, Nitrogen, and Argon

The Porta Cryo is a compact, portable cryogenic liquid cylinder designed for the safe storage, transport, and withdrawal of liquefied industrial gases such as Liquid Oxygen (LOX), Liquid Nitrogen (LIN), and Liquid Argon (LAR). Ideal for hospitals, laboratories, gas distribution, and on-site supply, the Porta Cryo enables efficient and controlled usage of cryogenic gases without the need for permanent infrastructure. CRYOCORP O2 LLP's Porta Cryo units are built with vacuum-insulated double-walled construction to maintain cryogenic temperatures and minimize product loss due to evaporation.



Each Porta Cryo unit from CRYOCORP O2 LLP undergoes vacuum testing, pressure trials, and performance validation before delivery. Optional accessories include vaporizer coils, filling connectors, and telemetry-based level indicators.

Technical Specifications (Example Range):

Model Capacity	180L / 240L / 265L / Custom sizes available
Operating Pressure	16 – 24 bar (adjustable depending on application)
Material	SS304 / SS316 inner vessel, SS or MS outer jacket
Insulation	Super vacuum with multi-layer insulation (MLI)
Evaporation Rate	≤ 2% per day (static holding condition)
Withdrawal Mode	Liquid withdrawal / Gas withdrawal
Mounted Options	Skid-mounted, wheeled base, or handle trolley style

Purpose & Functionality:

Porta Cryo cylinders serve as mobile dewars capable of holding cryogenic liquids at ultra-low temperatures. These vessels are equipped with self-pressurizing systems, withdrawal valves, and pressure-building coils, enabling easy dispensing in either liquid or gas form depending on application needs.

Applications:

- 1. Liquid Oxygen supply for hospitals and clinics
- 2. Liquid Nitrogen dispensing for food freezing and lab use
- 3. Portable liquid Argon for welding and metallurgy industries
- 4. Temporary supply during maintenance or plant shutdown
- 5. Mobile gas distribution in remote or high-demand zones

Key Features:

- 1. Vacuum-Insulated Stainless Steel Construction: Ensures long holding time and structural durability
- 2. Built-In Pressure Building System: Maintains optimal operating pressure for liquid withdrawal
- 3. Liquid and Gas Withdrawal Valves: Dual-mode use for maximum flexibility
- 4. Integrated Pressure Relief Valve & Burst Disc: For overpressure safety
- 5. Sturdy Mobility: Fitted with caster wheels or trolley handles for safe and easy transport
- 6. Level & Pressure Gauges: Real-time monitoring of content level and pressure

Why Choose CRYOCORP?

- 1. Eliminates dependency on high-pressure cylinders or pipeline networks
- 2. Low boil-off rate ensures long storage time and minimal gas loss
- 3. Safe, efficient, and mobile solution for cryogenic gas handling
- 4. Simple operation with rugged construction for field use
- 5. Available in standard or customized capacities for specific needs

09. Liquid Oxygen & Nitrogen Portable Tanks

Mobile Cryogenic Storage for On-Site & Field Applications

CRYOCORP O2 LLP's Liquid Oxygen (LOX) and Liquid Nitrogen (LIN) Portable Tanks are high-performance, vacuum-insulated containers designed for the safe storage, transportation, and on-demand supply of cryogenic gases in liquid form. Built for mobility and ease of use, these tanks are ideal for applications where a flexible, compact, and efficient gas supply is required. They are commonly used in medical oxygen delivery, industrial gas transport, field-based gas distribution, and temporary backup systems for oxygen and nitrogen supplies.



CRYOCORP O2 LLP provides full support with installation, commissioning, and optional vaporizer integration for continuous gas output. Portable tanks can be delivered with filling couplings, hose kits, and telemetry systems for remote monitoring.

Key Features:

- 1. Vacuum-Insulated Double Wall Construction: Ensures maximum thermal efficiency and minimal gas loss
- 2. Stainless Steel Inner Vessel: Corrosion-resistant, food- and medical-grade compliant
- 3. Built-In Pressure Building Coil: Maintains operating pressure for smooth withdrawal
- 4. Dual Withdrawal Modes: Liquid and gas withdrawal valves for versatile applications
- 5. Integrated Safety Systems: Pressure relief valves, burst discs, and check valves
- 6. Portable Design: Mounted on wheels or skids for easy movement and relocation
- 7. Level and Pressure Gauges: For real-time monitoring of tank conditions

Applications:

- 1. Emergency and mobile oxygen supply for hospitals and ambulances
- 2. Industrial gas usage in remote work sites
- 3. Liquid nitrogen for laboratories, food freezing, and inerting
- 4. Cylinder filling backup or mobile supply in temporary setups
- 5. Backup during plant maintenance or outages

Why Choose CRYOCORP?

- 1. Eliminates the need for cylinder clusters or fixed storage systems
- 2. Rapid deployment and easy transport
- 3. High safety margin with advanced pressure and relief controls
- 4. Customizable for application-specific use (medical, industrial, lab)
- 5. Built to ISO, BIS, and PESO standards with test certification

Technical Specifications (Example Range):

Tank Capacity	100L / 180L / 240L / 500L / Custom sizes
Working Pressure	16 – 24 bar (standard); up to 37 bar (optional)
Material	SS304/SS316 Inner Vessel; SS/MS Outer Jacket
Insulation Type	Vacuum with Multi-Layer Insulation (MLI)
Evaporation Rate	Wheeled base / Skid-mounted / Forklift slots

Purpose & Functionality:

These tanks maintain cryogenic temperatures (LOX: -183°C, LIN: -196°C) using advanced multi-layer vacuum insulation and are equipped with pressure-building and gas withdrawal systems, allowing for controlled gas delivery. Their mobile design makes them suitable for indoor or outdoor deployment with minimal infrastructure.



AI Tools for you



Savings Calculator

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